

Covariance-normal-ordered full Class-II energy composite

TECT verification-first repository · claim A7-CLASSII-RENORMALISED-ENERGY-COMPOSITE

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1. Purpose and scope

The preceding A6 packages fixed two facts. The full production Gaussian has a positive linear derivative contraction, and the fixed-floor geometric current K_A has a canonical common-even spectral lift. Neither fact defines $J_A K_A$, $|K_A|^2$, or an interacting Gibbs weight. This note closes the next, narrower question: the full Class-II *energy density as a Gaussian composite distribution*.

The result uses the fixed $L = 16$ torus, the hash-pinned A1 production quadratic symbol, three complex components (six real components), embedded Pauli generators, fixed

$$\varepsilon_\rho = 10^{-12} > 0,$$

one common real-even uniformly bounded scalar Fourier multiplier on every real component, pointwise convergence of that multiplier to one, compact support or a uniform Schwartz tail, and exact Fourier convolution or sufficient de-aliasing. The theorem is regulator independent only inside that class. The proof below uses regulator-independent Fourier majorants; it does not silently assume a pointwise kernel bound.

The result does not construct the interacting measure. In particular, L^2 convergence of an action is not uniform integrability of its exponential. The missing estimate is isolated in Section 8.

2. Exact finite-cutoff algebra

Realify the field as $X : \mathbb{T}_L^3 \rightarrow \mathbb{R}^6$ and each Hermitian generator as a real symmetric matrix S_A . Put

$$\rho = X^T X, \quad m_A = X^T S_A X, \quad q_A = \frac{m_A}{\rho + \varepsilon_\rho},$$

$$p_A(X) = 2S_A X, \quad v_A(X) = 2(S_A - q_A I)X.$$

For $Y_i = \partial_i X$, the currents are

$$J_{A,i} = p_A^T Y_i, \quad K_{A,i} = v_A^T Y_i.$$

Writing

$$Q_{\Pi} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} > 0,$$

define the symmetric six-real matrix

$$B(X) = \sum_A \left\{ a p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c v_A v_A^T \right\}.$$

Then the exact smooth cutoff density is

$$e_{\Pi,\Lambda}(x) = \frac{1}{2} \sum_{i=1}^3 Y_{\Lambda,i}(x)^T B(X_\Lambda(x)) Y_{\Lambda,i}(x). \quad (2.1)$$

This identity includes all J^2 , JK , and K^2 coefficients and is checked against the direct complex-current implementation.

Let

$$\Gamma_{\Lambda,i} = \mathbb{E}[\partial_i X_\Lambda(x) \partial_i X_\Lambda(x)^T] \quad (2.2)$$

be the *six-real* derivative covariance. Common evenness gives

$$\mathbb{E}[X_\Lambda(x) \partial_i X_\Lambda(x)^T] = 0. \quad (2.3)$$

The two jointly Gaussian vectors are therefore independent. The exact finite-cutoff conditional contraction is

$$\mathcal{C}_\Lambda(X) = \frac{1}{2} \sum_i \text{Tr}[B(X) \Gamma_{\Lambda,i}] = \mathbb{E}[e_{\text{II},\Lambda}(x) \mid X_\Lambda(x) = X]. \quad (2.4)$$

Equation (2.4), rather than only its leading asymptotic term, is the renormalisation condition.

In complex notation let

$$U_A = T_A \Psi, \quad V_A = (T_A - q_A I) \Psi, \quad D_{\Lambda,i} = \mathbb{E}[\partial_i \Psi_\Lambda \partial_i \Psi_\Lambda^\dagger].$$

Since $\Gamma_{\Lambda,i} = \frac{1}{2} \text{realify}(D_{\Lambda,i})$, (2.4) is equivalently

$$\mathcal{C}_\Lambda(\Psi) = \sum_{i,A} \left[a U_A^\dagger D_{\Lambda,i} U_A + 2b \text{Re}(U_A^\dagger D_{\Lambda,i} V_A) + c V_A^\dagger D_{\Lambda,i} V_A \right]. \quad (2.5)$$

The explicit one-half in the real covariance is load-bearing: inserting the complex $D_{\Lambda,i}$ directly into the real trace formula would double the subtraction.

Definition 1. For every generator and spatial direction define jointly

$$\begin{aligned} [J_{A,i}^2]_\Lambda^{\text{ren}} &= J_{A,i}^2 - p_A^T \Gamma_{\Lambda,i} p_A, \\ [J_{A,i} K_{A,i}]_\Lambda^{\text{ren}} &= J_{A,i} K_{A,i} - p_A^T \Gamma_{\Lambda,i} v_A, \\ [K_{A,i}^2]_\Lambda^{\text{ren}} &= K_{A,i}^2 - v_A^T \Gamma_{\Lambda,i} v_A. \end{aligned}$$

Their Q_{II} combination is

$$e_{\text{II},\Lambda}^{\text{ren}} = e_{\text{II},\Lambda} - \mathcal{C}_\Lambda(X_\Lambda). \quad (2.6)$$

3. Composite convergence theorem

Theorem 1 (fixed-floor covariance-normal closure). *For every $\kappa > 0$ the three products in Definition 1, and hence $e_{\text{II},\Lambda}^{\text{ren}}$, converge jointly as*

$$e_{\text{II},\Lambda}^{\text{ren}} \longrightarrow e_{\text{II}}^{\text{ren}} \quad \text{in} \quad L^2(\Omega; H^{-1-\kappa}(\mathbb{T}_L^3)). \quad (3.1)$$

The limit is common to every regulator in the class of Section 1. In particular,

$$\mathcal{V}_\Lambda^{\text{ren}} = \int_{\mathbb{T}_L^3} e_{\text{II},\Lambda}^{\text{ren}}(x) dx \longrightarrow \mathcal{V}^{\text{ren}} \quad \text{in} \quad L^2(\Omega), \quad (3.2)$$

and $\mathbb{E} \mathcal{V}^{\text{ren}} = 0$.

3.1 Gaussian regularity and coefficient bounds

The production symbol satisfies, uniformly at high momentum,

$$c\langle k \rangle^4 I \leq \mathcal{A}(k) \leq C\langle k \rangle^4 I. \quad (3.3)$$

Thus every real covariance entry is bounded by $C\langle k \rangle^{-4}$. The value field has regularity below one half, whereas a derivative has regularity below minus one half.

At fixed positive floor, q_A, p_A, v_A , and B are smooth functions of X . Since $|m_A| \leq \rho$, one has $|q_A| \leq 1$ and

$$|p_A(X)| + |v_A(X)| \leq C|X|, \quad |B(X)| \leq C|X|^2. \quad (3.4)$$

Every derivative of B has polynomial growth with a constant depending on the fixed floor. The point covariance converges, so all required Gaussian Sobolev norms of $B(X_\Lambda)$ are uniform in Λ .

3.2 Exact finite Gaussian-IBP covariance lemma

The rational coefficient is handled without an infinite Hermite truncation. We use the following finite lemma. Let $(X, Y, \tilde{X}, \tilde{Y})$ be jointly Gaussian finite-dimensional vectors with

$$\mathbb{E}[X \otimes Y] = 0, \quad \mathbb{E}[\tilde{X} \otimes \tilde{Y}] = 0,$$

and write

$$P = \max\{\|\mathbb{E}[Y \otimes \tilde{X}]\|, \|\mathbb{E}[X \otimes \tilde{Y}]\|\}, \quad Q = \|\mathbb{E}[Y \otimes \tilde{Y}]\|.$$

For symmetric matrix functions A, D with derivatives through order two of polynomial growth, put

$$\Phi_A = \frac{1}{2}A_{ab}(X)(Y_a Y_b - \mathbb{E}[Y_a Y_b]), \quad \tilde{\Phi}_D = \frac{1}{2}D_{cd}(\tilde{X})(\tilde{Y}_c \tilde{Y}_d - \mathbb{E}[\tilde{Y}_c \tilde{Y}_d]).$$

Uniformly over Gaussian value covariances in a compact set,

$$\|\mathbb{E}[\Phi_A \tilde{\Phi}_D]\| \leq C_{A,D}\{Q^2 + QP^2 + P^4\}. \quad (3.5)$$

Here is the exact tensor identity behind that bound. Put $P_{a\beta} = \mathbb{E}[Y_a \tilde{X}_\beta]$, $R_{c\alpha} = \mathbb{E}[\tilde{Y}_c X_\alpha]$, $Q_{ac} = \mathbb{E}[Y_a \tilde{Y}_c]$, $\Gamma = \mathbb{E}[Y \otimes Y]$, and $\tilde{\Gamma} = \mathbb{E}[\tilde{Y} \otimes \tilde{Y}]$. Then

$$\begin{aligned} & \mathbb{E}\{A_{ab}(X)(Y_a Y_b - \Gamma_{ab})D_{cd}(\tilde{X})(\tilde{Y}_c \tilde{Y}_d - \tilde{\Gamma}_{cd})\} \\ &= (Q_{ac}Q_{bd} + Q_{ad}Q_{bc})\mathbb{E}[A_{ab}(X)D_{cd}(\tilde{X})] \\ & \quad + \{P_{a\beta}(Q_{bc}R_{d\alpha} + Q_{bd}R_{c\alpha}) + P_{b\beta}(Q_{ac}R_{d\alpha} + Q_{ad}R_{c\alpha})\} \\ & \quad \times \mathbb{E}[(\partial_\alpha A_{ab})(X)(\partial_\beta D_{cd})(\tilde{X})] \\ & \quad + P_{a\beta}P_{b\gamma}R_{c\alpha}R_{d\delta}\mathbb{E}[(\partial_{\alpha\delta}A_{ab})(X)(\partial_{\beta\gamma}D_{cd})(\tilde{X})]. \end{aligned} \quad (3.5a)$$

To prove (3.5), apply Gaussian integration by parts successively to the four displayed derivative variables. The two conditional-centering terms cancel every same-vertex derivative pair. Each surviving derivative leg therefore either pairs with a derivative at the opposite vertex, contributing Q , or differentiates the opposite coefficient through a value-derivative covariance, contributing P . There are exactly three connection types:

$$(Q^2, P^0), \quad (Q^1, P^2), \quad (Q^0, P^4). \quad (3.6)$$

No coefficient is differentiated beyond order two. Hölder's inequality and the Gaussian Sobolev norms of A and D give (3.5). This is a finite identity and bound; it leaves no Hermite-series tail. Applied with $A = D = B$ it covers the complete rational coefficient, and applied to the three separate coefficient matrices it covers J^2 , JK , and K^2 jointly.

The same argument identifies the only divergent *local contraction in this Gaussian two-point composite construction*. Before centering, pairing the two derivative legs at one vertex gives

$$\sum_{|k| \lesssim \Lambda} |k|^2 \langle k \rangle^{-4} = \Theta(\Lambda), \quad (3.7)$$

and (2.4) removes it exactly while keeping B unexpanded. A same-point value-derivative contraction is zero by (2.3). We make no claim here about counterterms in an interacting perturbation expansion.

3.3 Bessel-weighted Fourier domination

Couple all cutoffs through the same underlying Gaussian Fourier modes. If $a_r(k) = \langle k \rangle^{-r}$, the production symbol and the common bounded scalar multipliers give, uniformly in both cutoffs,

$$|\hat{C}_{\Lambda M}(k)| \leq C a_4(k), \quad |\hat{P}_{\Lambda M}(k)| \leq C a_3(k), \quad |\hat{Q}_{\Lambda M}(k)| \leq C a_2(k). \quad (3.8)$$

The first sequence is summable in dimension three, so the value covariances converge uniformly. Polynomial growth and uniform Gaussian moments then make all coefficient expectations in (3.5a) bounded and continuous under a change of cutoff.

The needed discrete convolutions are, for every $\eta > 0$,

$$a_2 * a_2 \lesssim a_1, \quad a_2 * a_3 * a_3 \lesssim_\eta a_{2-\eta}, \quad a_3 * a_3 * a_3 * a_3 \lesssim_\eta a_{3-\eta}. \quad (3.9)$$

They follow by splitting each lattice sum into the regions where one input frequency is at most half the output frequency and its complement. The first estimate is sharp. It corresponds to the two derivative-derivative bridges; the other two are the QPR and P^2R^2 lines of (3.5a).

The executable connection-pattern audit enumerates exactly these three families rather than declaring a hand-written multivertex graph table. Its frozen-gradient variance sum is only a numerical check of the worst Q^2 scaling; the proof of the full rational coefficient is the exact identity (3.5a) together with (3.8)–(3.9).

3.4 Sobolev Cauchy convergence and scheme independence

Put $s = 1 + \kappa$ and let G_s be the positive periodic Bessel kernel with Fourier coefficients $\langle k \rangle^{-2s}$. Positivity follows from its heat-kernel representation. From (3.9),

$$\int G_s(z) |Q_{\Lambda M}(z)|^2 dz \lesssim \sum_k \langle k \rangle^{-2s-1} < \infty \iff s > 1. \quad (3.10)$$

The P^2R^2 term is strictly better by the last estimate in (3.9), and the QPR term is bounded by Cauchy–Schwarz in the positive measure $G_s(z) dz$. Thus (3.5a) is uniformly integrable against G_s for every $s > 1$.

Pointwise multiplier convergence and the same absolutely summable convolution majorants imply convergence of each G_s -weighted covariance term. Uniform convergence of the value covariance handles the coefficient expectations in (3.5a): truncate the Gaussian values at fixed radius, use continuity in the finite-dimensional covariance there, and then remove the truncation with the uniform polynomial moments.

For $R_{\Lambda M} = e_{\Pi, \Lambda}^{\text{ren}} - e_{\Pi, M}^{\text{ren}}$, stationarity and Parseval give

$$\mathbb{E} \|R_{\Lambda M}\|_{H^{-s}}^2 = \int_{\mathbb{T}_L^3} G_s(z) \mathbb{E}[R_{\Lambda M}(z) R_{\Lambda M}(0)] dz. \quad (3.11)$$

Polarising the four cross-cutoff covariances and applying the preceding dominated-convergence argument makes (3.11) tend to zero. This proves the common Cauchy limit in $L^2(\Omega; H^{-1-\kappa})$ for every $\kappa > 0$. At

$s = 1$ the sharp Q^2 sum in (3.10) is logarithmic, so the H^{-1} endpoint is not claimed. Pairing with the constant test function gives (3.2), and conditional centering in (2.4) gives zero mean at every cutoff and in the limit.

This direct Fourier–Wick proof is the pinned specialisation used here. The general BPHZ framework and stochastic model convergence are developed by Chandra and Hairer, <https://arxiv.org/abs/1612.08138>; the present proof does not import a black-box interacting-measure conclusion from that work.

4. Leading contraction and exact finite scheme

For the sharp cube, spatial symmetry makes all complex derivative covariances equal to the A6 matrix D_N . The A6 limit gives

$$\frac{D_N}{N} \longrightarrow \delta_{\text{cube}} I_3, \quad \delta_{\text{cube}} = 0.01619898645075695. \quad (4.1)$$

Substituting (4.1) into (2.5) recovers

$$\frac{\mathcal{C}_N(\Psi)}{N} \longrightarrow \delta_{\text{cube}} W_\varepsilon(\Psi), \quad (4.2)$$

where, with $s = |\Psi_1|^2 + |\Psi_2|^2$,

$$W_\varepsilon = 9(a + 2b + c)s - \frac{6bs^2}{\rho + \varepsilon_\rho} - \frac{3cs^2(\rho + 2\varepsilon_\rho)}{(\rho + \varepsilon_\rho)^2}. \quad (4.3)$$

Equation (4.2) is an asymptotic check, not the definition. Exact $D_{\Lambda,i}$ subtraction removes the finite regulator-dependent part and pins one covariance-normal renormalisation condition.

5. Frozen-background determinant check

Fix a deterministic background u and freeze $B(X)$ at $B(u)$. Relative to the Gaussian quadratic operator, the derivative perturbation defines a positive operator $T_{u,\Lambda}$ whose high-mode eigenvalues are $O(|k|^{-2})$. Hence

$$\sup_{\Lambda} \|T_{u,\Lambda}\|_{\text{HS}}^2 \lesssim \sum_k \langle k \rangle^{-4} < \infty. \quad (5.1)$$

The exactly centered quadratic partition factor is

$$\mathbb{E} \exp \left\{ -\frac{1}{2} \langle X, L_u X \rangle + \frac{1}{2} \text{Tr } T_{u,\Lambda} \right\} = \det_2(I + T_{u,\Lambda})^{-1/2}, \quad (5.2)$$

and

$$0 \leq \frac{1}{2} \sum_j [\lambda_j - \log(1 + \lambda_j)] \leq \frac{1}{4} \sum_j \lambda_j^2. \quad (5.3)$$

This proves a convergent frozen-coefficient determinant and checks that the remaining diagrams are Hilbert–Schmidt. It is not an estimate for the self-coupled random coefficient $B(X)$ and is not used to claim a Gibbs measure.

6. Sharp pointwise running-mass threshold

At the production coefficients,

$$a = 0.0045, \quad b = 0.001875, \quad c = 0.00234375.$$

Put

$$h = 9(a + 2b + c) = 0.09534375, \quad d = 6b + 3c = 0.01828125. \quad (6.1)$$

Equation (4.3) gives the exact identity

$$hs - W_\varepsilon = \frac{6bs^2}{\rho + \varepsilon_\rho} + \frac{3cs^2(\rho + 2\varepsilon_\rho)}{(\rho + \varepsilon_\rho)^2} \geq 0. \quad (6.2)$$

Moreover $W_\varepsilon/s \rightarrow h$ as $s/\rho \rightarrow 0$ with a nonzero third component. Therefore

$$\sup_{\Psi: s > 0} \frac{W_\varepsilon(\Psi)}{s} = h. \quad (6.3)$$

For the leading subtraction $-t_N W_\varepsilon$, a pointwise family-mass repair $t_N m s$ is nonnegative for every field exactly when $m \geq h$. Since the production family energy is $\frac{1}{2} \Delta \mu_N s$, the leading sharp shift is

$$\Delta \mu_N \geq 2h \delta_{\text{cube}} N + O(1) = 0.00308894422882794 N + O(1). \quad (6.4)$$

If $m < h$, choose a fixed small orientation $x = s/\rho$ such that

$$A_x = (h - m)x - dx^2 > 0.$$

A homogeneous amplitude then has asymptotic potential

$$\frac{\gamma}{6} \rho^3 - t_N A_x \rho + O(\rho^2),$$

whose minimum is

$$-\frac{2^{3/2}}{3\sqrt{\gamma}} (t_N A_x)^{3/2} + O(t_N). \quad (6.5)$$

Thus every subcritical pointwise mass repair retains the raw $-\Theta(N^{3/2})$ escape. At the sharp value, (6.2) produces a divergent positive orientation penalty and can collapse the first doublet under additional uniform-energy and tightness assumptions. This explains why the pointwise-coercive repair is not adopted as the physical nondegenerate renormalisation condition.

7. Bare-route negative control

The zero set of the conditional contraction is not the zero set of the pathwise Class-II energy. For

$$\Psi(x) = e^{ik \cdot x} u \quad (7.1)$$

with constant $u \in \mathbb{C}^3$, both ρ and every m_A are constant. Therefore $J_A = \nabla m_A = 0$ and $K_A = 0$, even if the first doublet of u is nonzero. But (4.3) gives $W_\varepsilon(u) > 0$ for such a field.

More generally, positive definiteness of Q_{II} implies that $F_{\text{II}} = 0$ exactly when every J_A and K_A vanishes. This includes fixed-density, fixed-Bloch-direction phase branches, not only the pure third component. Consequently the conditional Gaussian mean cannot be used as a deterministic rate function for the bare Gibbs law. Any bare-concentration proof must compare the entropy and free energy of all null branches.

8. Interacting-measure boundary

Let U_Λ denote the local quartic–sextic part relative to the production Gaussian. Since

$$\mathbb{E}\mathcal{V}_\Lambda^{\text{ren}} = 0$$

and the Gaussian local moments are uniformly bounded, Jensen gives a uniform positive lower bound on the finite-cutoff partition functions:

$$Z_\Lambda = \mathbb{E} \exp[-U_\Lambda - \mathcal{V}_\Lambda^{\text{ren}}] \geq \exp[-\mathbb{E}U_\Lambda] > z_* > 0. \quad (8.1)$$

The positive sextic gives finite-dimensional normalisability. Neither fact gives uniform integrability of the weights.

The missing constructive estimate is

$$\sup_\Lambda \mathbb{E} \exp\{-p[U_\Lambda + \mathcal{V}_\Lambda^{\text{ren}}]\} < \infty \quad \text{for some } p > 1. \quad (8.2)$$

If (8.2) is proved, (3.2), Vitali convergence, and the common-Gaussian construction yield L^1 convergence of densities, partition convergence, tightness, a full-sequence finite-volume Gibbs limit, and convergence of smeared correlations. This is the named **A7-CLASSII-NELSON-EXPONENTIAL-BOUND** gate. It is not discharged in this note.

9. Executable audit

The primary implementation reconstructs the exact tensor counterterm, real/complex factor, conditional Gaussian mean, three regulator Hilbert–Schmidt tails, the finite Gaussian-IBP connection patterns, mass threshold, plane-wave negative control, and frozen determinant. It passes 29/29.

The non-importing implementation uses direct mode enumeration, independently coded generators and coefficients, a separate Monte Carlo stream, direct variance sums, a recursive connection-pattern enumeration, the actual three-dimensional parity matrix, and a second determinant calculation. It passes 17/17. The integrated verifier performs all authority hashes before execution and adds 28 cross-route assertions, for 74/74 in total.

The principal derived diagnostics are

exact counterterm / N error at N=128	0.00939344390845
variance-tail slopes (cube, ball, smooth)	-1.11272, -1.10545, -1.08903
conditional Monte Carlo z-score	0.176135
h = sup W/s	0.09534375
2 h delta_cube	0.00308894422883
plane-wave max J , K	2.22045e-16
plane-wave W	0.0534857142857
frozen determinant remainder at N=64	0.000102756574758
corresponding half-HS upper bound	0.000105206016708

Reproduce the package with

```
python codes/foundations/a7_classii_renormalised_energy_verify.py
```

10. Devil’s-advocate

1. **DISMISSED:** the A6 complex derivative covariance can be inserted unchanged into the six-real trace. It cannot; the real covariance is one half of the realification. Both implementations test the factor.

2. **VALID-with-mitigation:** subtracting only $\delta_{\text{cube}}NW_\varepsilon$ can leave a finite scheme term. The definition uses each regulator's exact $\Gamma_{\Lambda,i}$ and thereby fixes the covariance-normal finite renormalisation condition.
3. **DISMISSED in the Gaussian-composite scope:** the rational coefficient creates an uncontrolled Hermite tail. Exact identity (3.5a) acts directly on B and uses only its first two derivatives. Conditional centering removes the local derivative pair, evenness removes the local mixed pair, and the three surviving two-point connection families are controlled by (3.9). This does not assert convergence of an interacting perturbation series.
4. **VALID-with-mitigation:** arbitrary component-split or asymmetric regulators may carry a first-level or area anomaly. They are excluded. The theorem inherits the full common real-even scalar multiplier and exact/dealiased-product boundary of A6.
5. **UPHELD:** composite L^2 convergence is not a constructive Gibbs theorem. Uniform negative-exponential control (8.2) is separately named and remains open.
6. **VALID-with-mitigation:** the frozen-background determinant might be mistaken for the self-coupled measure. Section 5 states and uses it only for deterministic $B(u)$; it is an audit of the Hilbert–Schmidt remainder.
7. **UPHELD:** a pointwise running mass at the sharp threshold can produce a degenerate first-doublet limit. The result is recorded as a stability boundary, not selected as the physical prescription.
8. **UPHELD:** $W_\varepsilon^{-1}(0)$ does not classify the bare pathwise zero set. The exact plane-wave counterexample prevents a pure-third concentration overclaim.
9. **VALID-with-mitigation:** constants can blow up as $\varepsilon_\rho \downarrow 0$. The floor is fixed; no uniform floor-removal statement is made.

11. Result footer

Result ID:	A7-CLASSII-RENORMALISED-ENERGY-COMPOSITE
Precise statement:	Exact covariance-normal ordering jointly defines J^2 , J^*K , K^2 and the full Class-II energy density; the common-even fixed-floor spectral sequence converges in $L^2(\Omega; H^{(-1-\kappa)})$.
Scope:	$L=16$ production q^{-4} Gaussian; three complex fields; six-real convention; $\text{eps_rho}=1\text{e-}12$; common bounded real-even scalar multipliers with compact support or uniform Schwartz tails; exact/dealiased products.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION; A6-CLASSII-UV-POWER-COUNTING; A6-CLASSII-K-COMPOSITE-DEFINITION.
Evidence:	exact Gaussian-IBP/Fourier proof; exact algebra; primary, non-importing independent, and integrated verifier.
Falsification gate:	wrong real/complex factor; missing Gaussian-IBP connection; non-Cauchy centered variance; failed scheme/source/counterterm/determinant assertion.
Tier before / after:	new / T5 PINNED-CLOSURE@FIXED-FLOOR-COVARIANCE-NORMAL-ENERGY-COMPOSITE.
No-overclaim statement:	No uniform exponential bound, interacting Gibbs measure, full-field bare concentration, floor removal, arbitrary regulator universality, infinite volume, phase transition, BCC, T6, or T7.
Next required action:	Prove A7-CLASSII-NELSON-EXPONENTIAL-BOUND; keep the bare concentration branch separate.